

Input

Representations

Model families

A. SVM-combined

B. LSTM-combined

C. BiLSTM-MHA

Sequence-index encoding

indices 1–20
pad/truncate to $L = 50$

GPSD descriptors

7 physicochemical groups
147 initial $\rightarrow$ 126
selected features

Sequence indices +
GPSD

Feature concatenation

RBF-SVM classifier

$C = 1.0$
 $\gamma = \text{scale}$

AMP/non-AMP

Sequence branch
indices ($L = 50$)

Embedding
128-d
+ dropout 0.3

LSTM
representation

GPSD branch
126-d
descriptors

Dense
descriptor
branch

Concatenate

Dense + ReLU

Sigmoid output
threshold = 0.5

AMP/non-AMP

Sequence branch
indices ($L = 50$)

Embedding
128-d
+ dropout 0.3

BiLSTM
context

Multi-head
attention

GPSD branch
126-d
descriptors

Dense
descriptor
branch

Concatenate

Dense + ReLU

Sigmoid output
threshold = 0.5

AMP/non-AMP

Peptide
sequence input

20 standard
amino acids

0 = padding
symbol

- Sequence arrays use 20 standard amino-acid symbols; 0 denotes padding.
- GPSD contributes 126 selected physicochemical descriptors after feature selection.
- Learned preprocessing objects are fitted on the model-development partition.